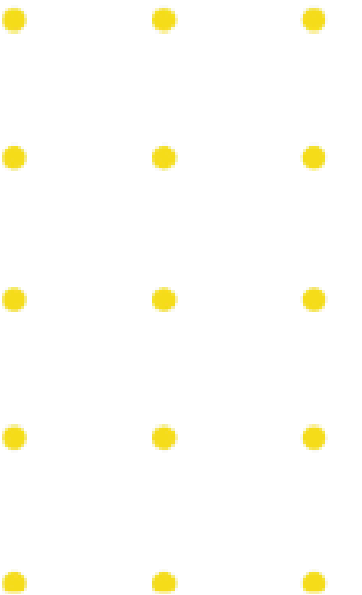
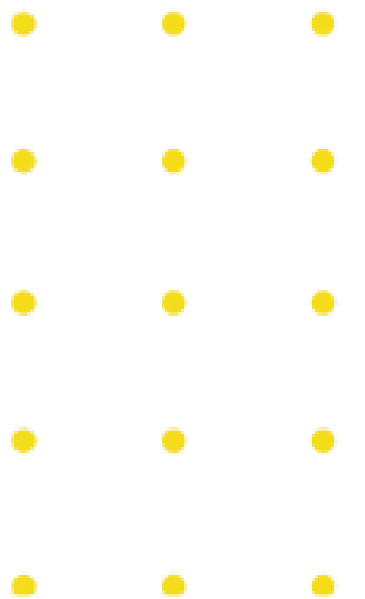


Module 5:

Networking Devices

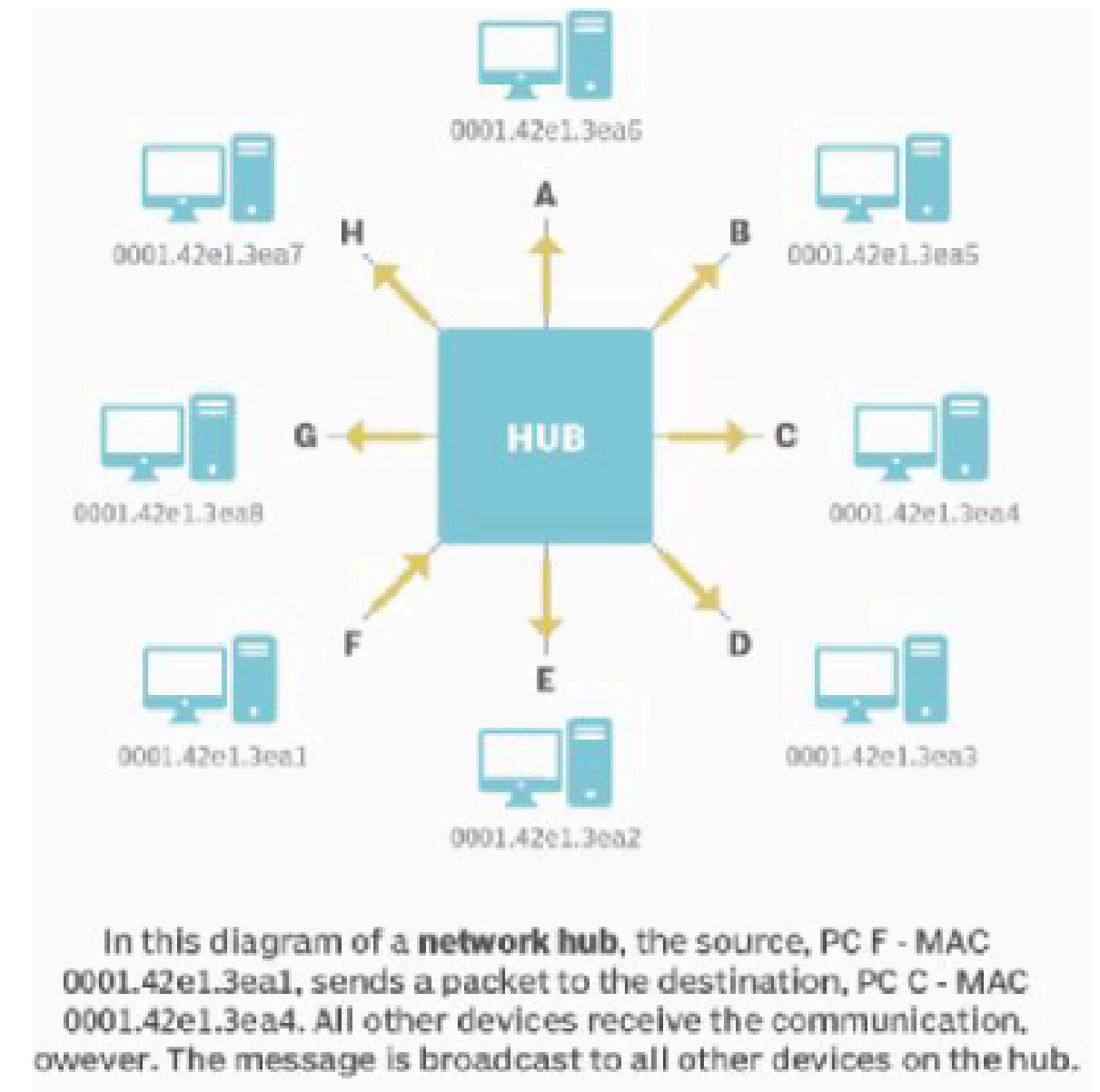


Topic 3: Hubs/Switches/Routers



HUBs

- A network hub is simply a connection node that distribute data to every computer or Ethernet-based device connected to it
- Hub is less complex than Switches.
- Network Hubs are in layer one in OSI Model
- There are two types of hubs
 - Active Hubs: strengthen signals.
 - Passive: works only as a node of connection

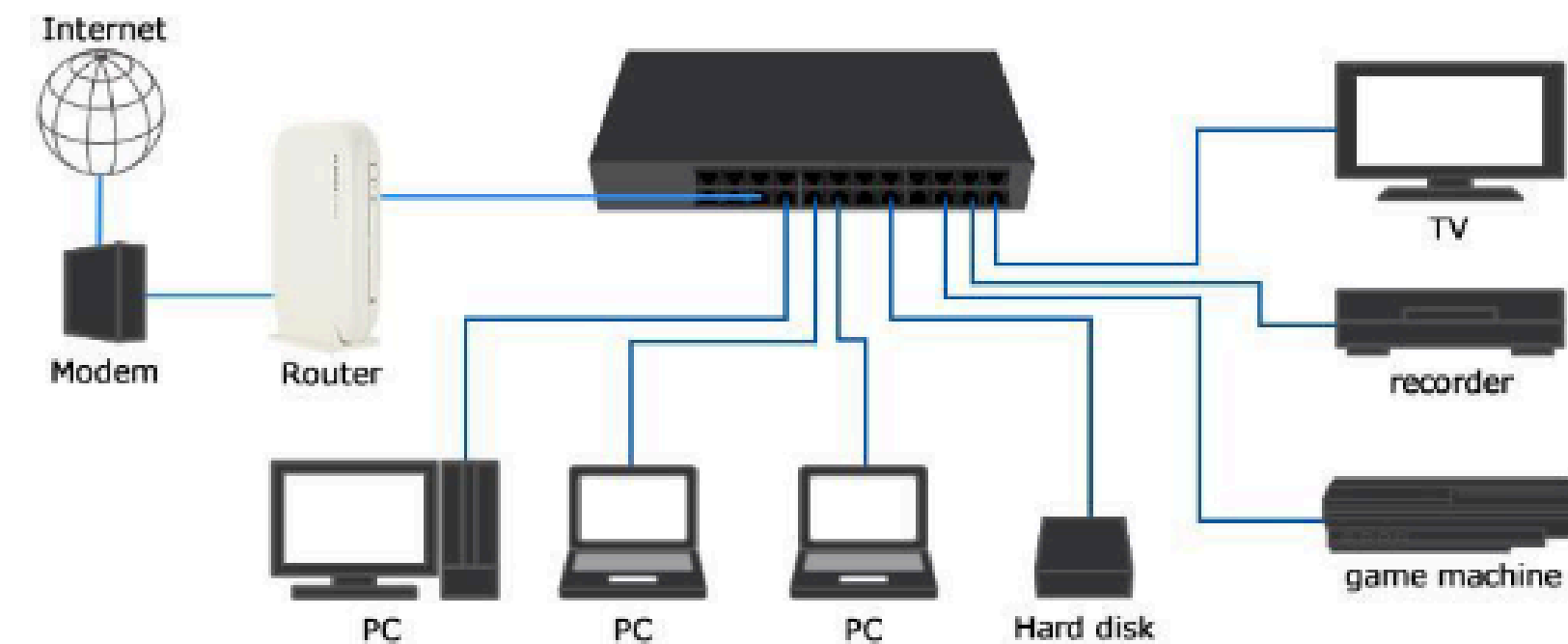


HUBs

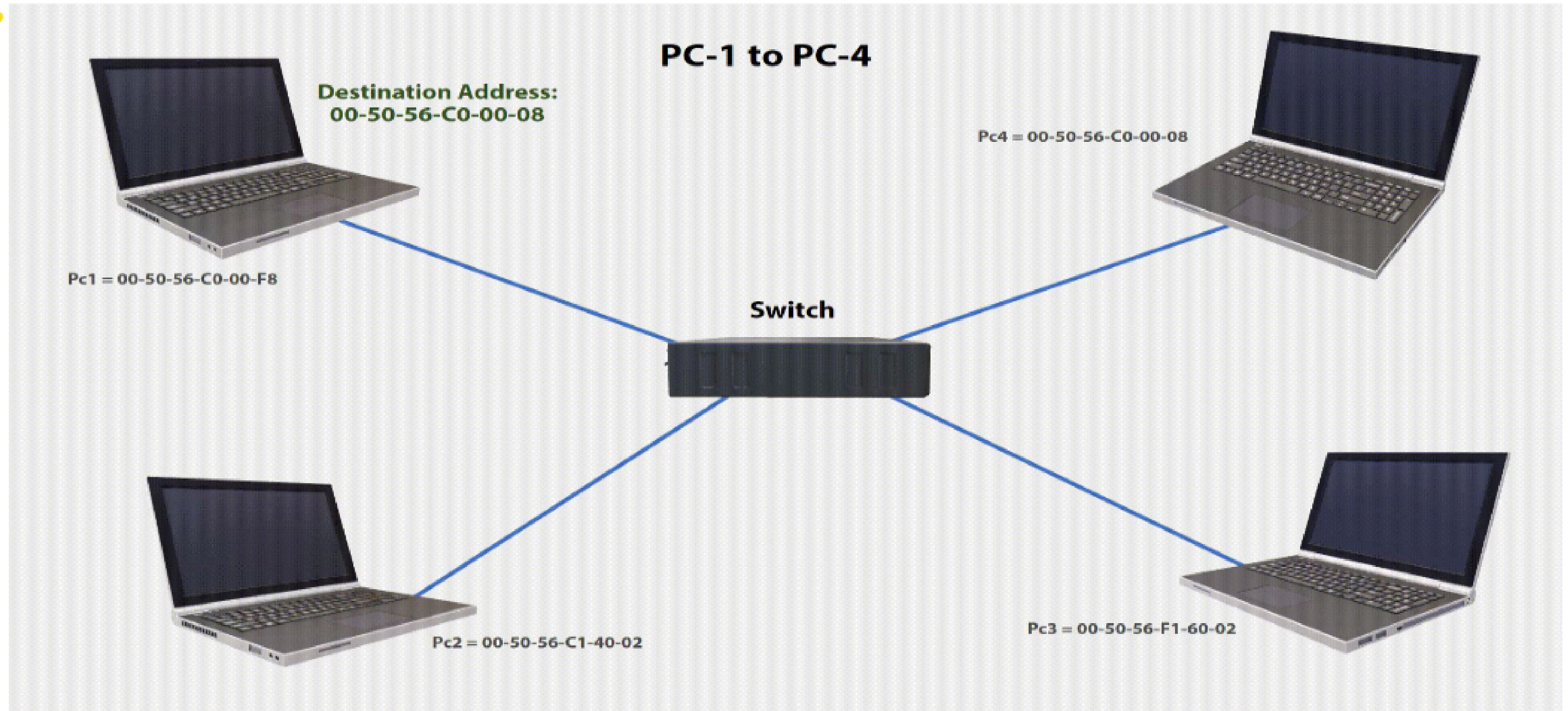


Switches

- **Network Switches** connect devices directly through their physical ports.
- It operates at Data Link Layer in OSI Model.
- Its function: packet switching among connected devices.
- Uses MAC address to identify source and destination of packets (frames) being transferred.
- It is used to construct Virtual Local Area Networks VLANs
- switch(es) supporting VLAN capabilities can be configured to define multiple Virtual LANS over single physical LAN infrastructure.



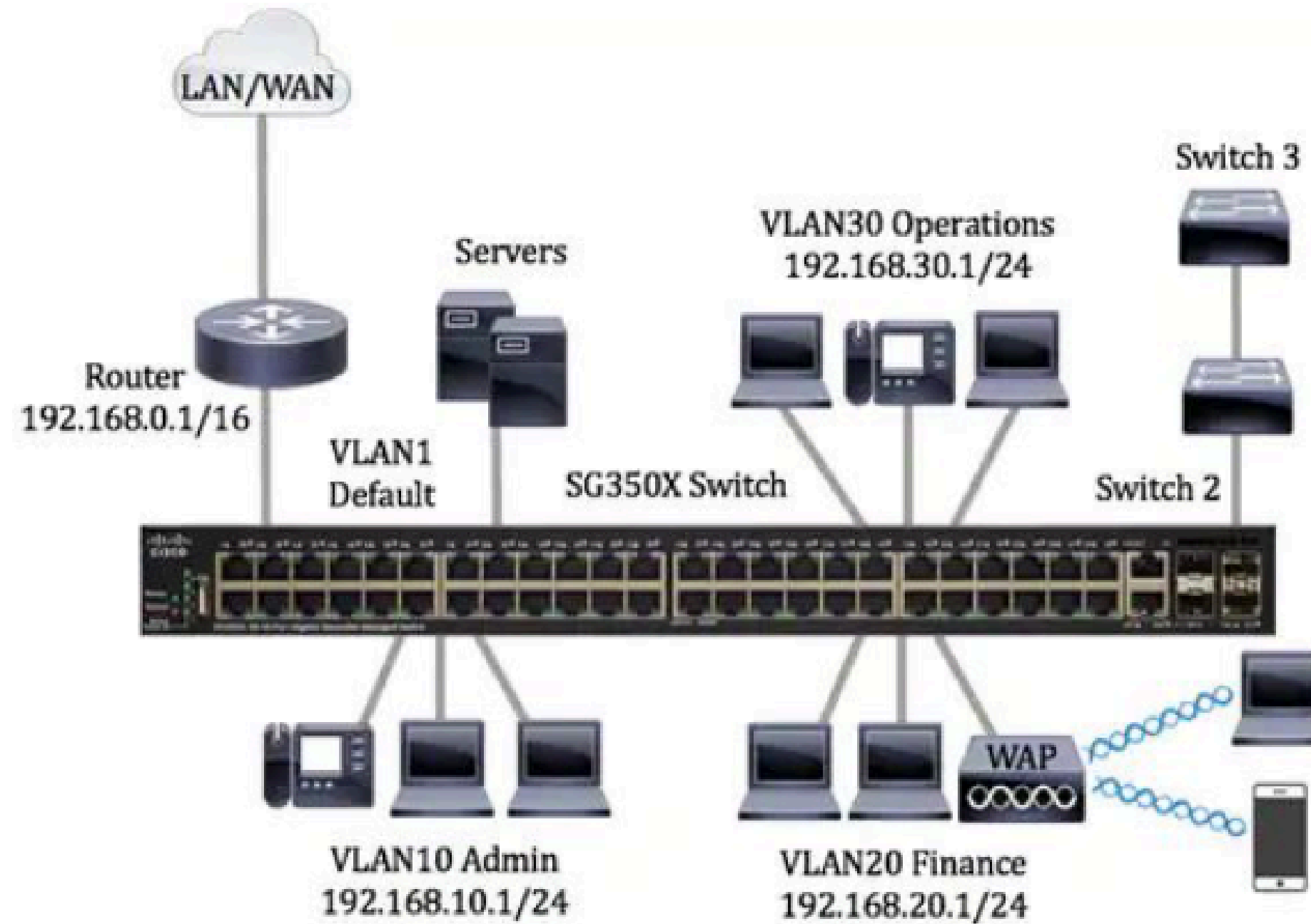
Switches



Switches: VLANs

- **VLANs** are a group of computers under the same broadcast domain.
- VLANs could be configured using **Switches** by placing some interfaces into one broadcast domain and some interfaces into another. Each VLAN acts as a subgroup of the switch ports in an Ethernet LAN.
- A VLAN acts like a physical LAN, but it allows hosts to be grouped together in the same broadcast domain even if they are not connected to the same switch.

Switches: VLANs



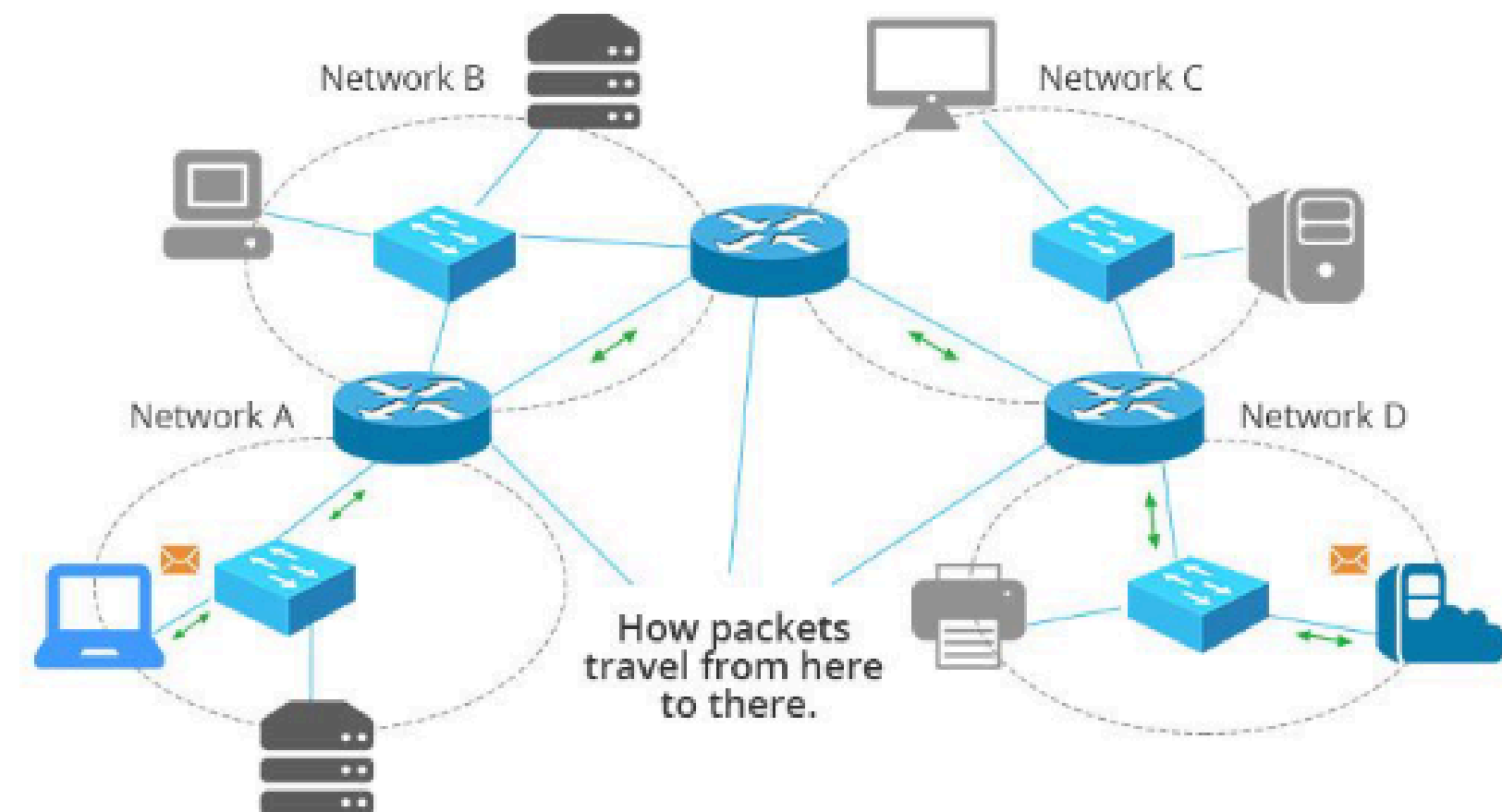
Source: <https://www.cisco.com/c/en/us/support/docs/smb/switches/cisco-small-business-300-series-managed-switches/smb5653-configure-port-to-vlan-interface-settings-on-a-switch-through.html>

Switches: VLANs

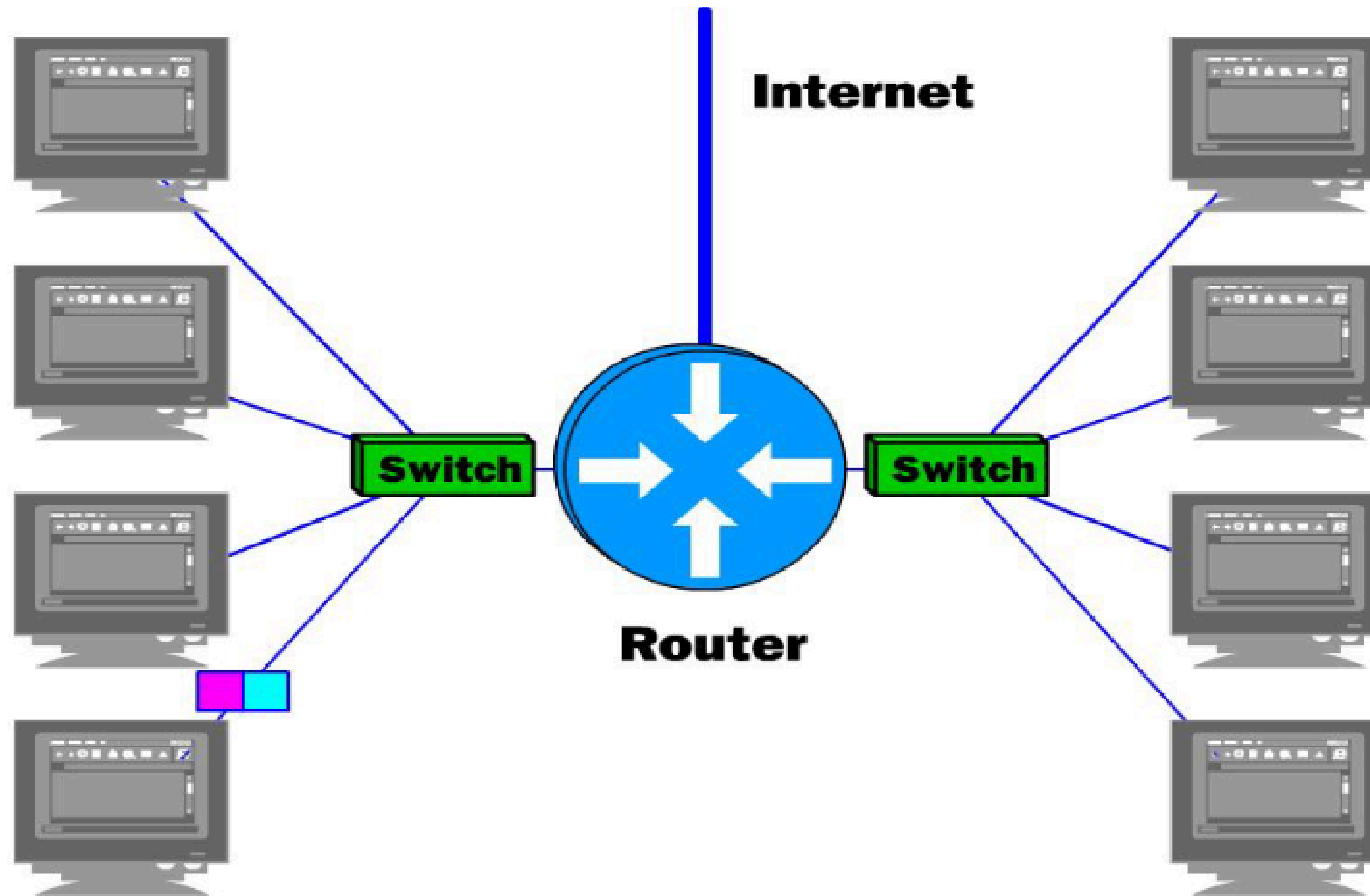
- a VLAN could be configured using **multiple switches** but all computers in an arbitrary VLAN are treated as its own subnet or broadcast domain.
- Here comes the advantage! VLANs decrease the number of hosts that receive copies of frames which reduce security risks.
- You can create network that groups a department in company without the limitation of position.
- Decrease the size of broadcast for each VLAN by increasing **broadcast domains**.

Routers

- Routers has a main function to transfer packets from a sending host (client/server) to receiving host (client/server).
- Routers function is separated into two techniques.
 - **Forwarding**; where incoming packets into a router input link should send on one of routers output link; this happens within the router.
 - **Routing**: at network level, packets route should be determined along its trip from the **sending host** to the **receiving host** passing through intermediate routers that direct packets to the **correct path**.



Routers



Routers Vs Switches

Routers

compute tables using routing algorithms, IP addresses

Can work within both wired and wireless network situations.

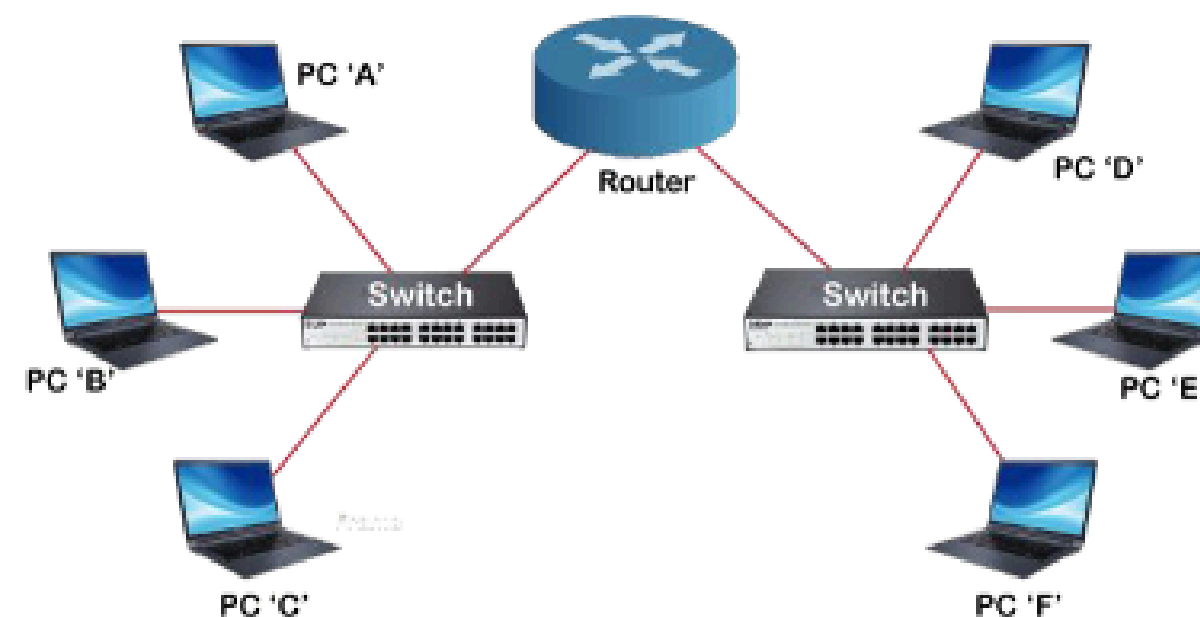
It connects multiple switches & their corresponding networks.

Switches

learn forwarding table using flooding, learning, MAC addresses

Restricted to wired network connections.

It connects multiple networked devices in the network.



Connection of networks through Router